

Institute of Actuaries of India

Subject CB2-Business Economics

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u>	A	[1.5 Marks]
<u>Solution 2:</u>	C	[1.5 Marks]
<u>Solution 3:</u>	C	[1.5 Marks]
<u>Solution 4:</u>	A	[1.5 Marks]
<u>Solution 5:</u>	A	[1.5 Marks]
<u>Solution 6:</u>	B	[1.5 Marks]
<u>Solution 7:</u>	A	[1.5 Marks]
<u>Solution 8:</u>	D	[1.5 Marks]
<u>Solution 9:</u>	D	[1.5 Marks]
<u>Solution 10:</u>	D	[1.5 Marks]
<u>Solution 11:</u>	A	[1.5 Marks]
<u>Solution 12:</u>	D	[1.5 Marks]
<u>Solution 13:</u>	C	[1.5 Marks]
<u>Solution 14:</u>	B	[1.5 Marks]
<u>Solution 15:</u>	D	[1.5 Marks]
<u>Solution 16:</u>	C	[1.5 Marks]
<u>Solution 17:</u>	C	[1.5 Marks]
<u>Solution 18:</u>	C	[1.5 Marks]
<u>Solution 19:</u>	C	[1.5 Marks]
<u>Solution 20:</u>	B	[1.5 Marks]
<u>Solution 21:</u>	D	[1.5 Marks]
<u>Solution 22:</u>	A	[1.5 Marks]
<u>Solution 23:</u>	A	[1.5 Marks]
<u>Solution 24:</u>	B	[1.5 Marks]
<u>Solution 25:</u>	A	[1.5 Marks]
<u>Solution 26:</u>	D	[1.5 Marks]

Solution 27: The Government can intervene in the free-market in the following manner:

- Price Control: The government can ask the firms to charge a price which is at a social optimum level. It can prevent the exploitation of consumers [1]
- The government can have price floors or price ceilings. Price floors could cause government buy up excess supply which is expensive and it at the cost of taxpayers' money [1]
- Government can use its talented resources like civil servants, economists, lawyers, and other technocrats in developing policy for the markets which maximises benefits rather than market forces taking their course [1]
- Government has sometimes more information than the individuals and firms combined and hence can take better decision to protect the consumer welfare. E.G if the Government knows before the public about the incoming pandemic (before the public) it can manage the resources such as increasing the production and reducing the price of life saving drugs [1]
- Governments can alter the taxes in a manner to cure the regional imbalance in the country. This would help reduce congestion and pollution hence leading to lower social costs [1]

[5 Marks]

Solution 28:

- i) 5000 per quintal would mean 50 per KG

$$Q_s = 7(P - 50)$$

$$Q_s = 7P - 350 \text{ Here } P \text{ is the new market price}$$

Here P is the new market price per KG

[1]

- ii) The equilibrium quantity demanded before

$$Q_s = Q_d$$

$$7P = 3600 - 5P$$

$$P = 300 \text{ per KG}$$

$$Q_d = 2100 \text{ KG}$$

[1]

New quantity demanded

$$7P - 350 = 3600 - 5P$$

$$12P = 3950$$

$$P = 329 \text{ per KG}$$

$$Q_d = 1954 \text{ KG}$$

[1]

The quantity demanded now versus previous $1954/2100 = 93\%$

The reduction in quantity demanded is 7%

[1]

[3]

- iii) Tax revenue shall be $50 \times 1954 = \text{INR } 97700$

[1]

- iv) From the above the new equilibrium price is INR 329 per KG. The price has risen by INR 29 because of the tax.

The consumer is bearing $29/50$ that is 58% of the tax.

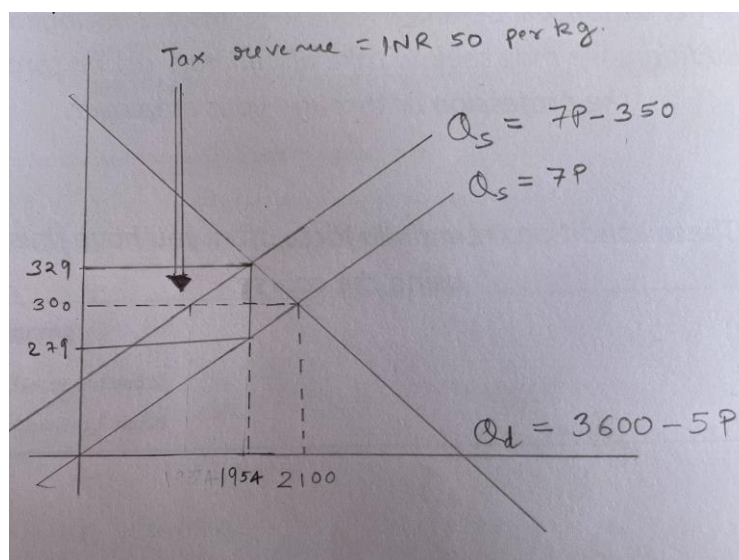
[1]

The producer is bearing $21/50$ that is 42% of the tax

[1]

[2]

- v)



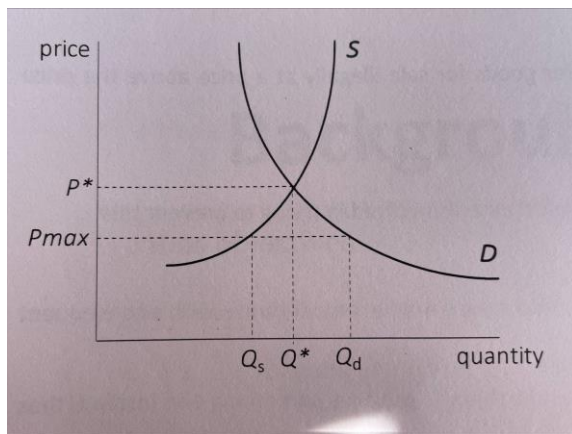
[3]

[10 Marks]

Solution 29:

A price ceiling is where government makes it illegal for to charge more than a certain amount for the good

[0.5]



[3]

The diagram shows the situation in which a price ceiling is set a price level below the free-market equilibrium price.

Advantages

- The lower price means that some people will be able to afford hand sanitisers who could not do so under free market equilibrium [0.5]
- Otherwise, due to a huge demand the hand sanitisers would become unaffordable for certain sections of the society [0.5]

Disadvantages

- With a price ceiling of P_{max} suppliers will want to supply a quantity Q_s whereas the consumers would demand a quantity Q_d given the price ceiling is implemented. [1]
- Hence there would be a shortage of $Q_d - Q_s$. A price ceiling set below the free market price therefore produces shortages [0.5]
- This will lead to consumer demand not getting met. Since this is an essential item consumers demand not getting met would lead to bad consequences. [0.5]
- This will increase the pressure on the government to use taxpayers' money to fund the shortage of supply [0.5]
- The suppliers may get discouraged and exit the market leading to further shortage [0.5]

[Max 7 Marks]

Solution 30: Total utility is the total satisfaction derived from the consumption of goods within a given time period... [0.5]

...whereas Marginal utility is the addition to total satisfaction derived from the consumption of one extra unit of a good within a given time period, assuming that the consumption of other goods is held constant. [0.5]

The total utility derived from rice is high but for most of us the marginal utility of rice is very low. [0.5]

We can consume nearly as much rice as we want so extra utility derived by extra spoon of rice is very low. [0.5]

In contrast the total utility derived from diamonds is much lower than derived from rice because we consume so few of them. [0.5]

This would mean the marginal utility of diamonds is much higher than that of rice.	[0.5]
It's not the total utility but the marginal utility that determines the price.	[0.5]
Higher marginal utility diamonds lead to its higher price.	[0.5]
If we were in a famine then the marginal utility of rice would be huge and we would be prepared to spend a huge amount of money for rice.	[1]
Finally, price is also dependent on demand and supply.	[0.5]
The supply of diamonds is very limited whereas the supply of rice is significantly high. Hence a lower price of rice.	[1]
[Max 5 Marks]	

Solution 31:

i)	This is not a perfectly contestable market.	[0.5]
	To build flats a large amount of capital is needed which is not available with everyone. This requires a lot of expertise which may not be available with everyone.	[1]
	This also requires approvals from the government which may be difficult to get.	[0.5]
	Finally, this market has a brand loyalty. Because a person investing a huge amount to purchase flat may go for reliable brands and not the newcomers.	[1]
[Max 2]		
ii)	This is a perfectly contestable market	[0.5]
	To open a Grocery store you need a very little investment	[0.5]
	There are no regulations or government permissions required.	[0.5]
	There is no brand loyalty as people are more lured by the brands you keep in stock than your brand.	[0.5]
	The customers will switch to another Grocery store if your prices are higher	[0.5]
[Max 2]		
iii)	This is not a perfectly contestable market.	[0.5]
	To operate an airline a large amount of capital is needed which is not available with everyone. To operate an airline, you need to hire pilots which may be in very short supply.	
	Hence you need to pay significantly more to poach them	[1]
	This requires aircrafts which are not purchased off the shelf.	
	The time between order and delivery is huge and scaling up becomes very difficult	[1]
[Max 2]		
[6 Marks]		

Solution 32:

Neo Keynesians attempted to explain why according to theory equilibrium may not be reached in markets.	[0.5]
They offered explanation in terms of imperfections in competitive markets and factors such as (wage rigidity) which could not result in prices interest rates and wages remaining at a level incompatible with those needed to establish equilibrium.	[1]

Neo Keynesians argued that if full employment was not naturally achieved, government could intervene to ensure full employment. [0.5]

Drawbacks

Models built on the basis of this theory do not allow for changes in economic environment and Government policy or sometime unpredictable behaviour of economic agents. [1]

In the 70s the appearance of stagflation a combination of high unemployment and high inflation question the validity of Neo Keynesian ideas [1]

[Max 4 Marks]

Solution 33:

i) Equilibrium occurs where Aggregate Expenditure AE = National Income Y

The equilibrium level of national income is given by:

$$Y = C + I + G + X - M$$

$$= 100 + .6Y_d + 100 + 200 + 40 - (40 + 0.1Y)$$

$$= 100 + .6 * (1 - t) * Y + 100 + 200 + 40 - (40 + 0.1Y)$$

$$= 400 + .6 * (1 - .5) * Y - .1Y$$

$$= 400 + .2Y$$

=>

$$.8Y = 400$$

$$Y = 500$$

[3]

ii) $C = 100 + 0.6(1 - 0.5)Y = 100 + .6 * .5 * 500 = 250$ [1]

iii) Trade deficit = (Import – Export) = $40 + .1 * 500 - 40 = 50$ [1]

[5 Marks]

Solution 34:

i) **Hysteresis**

Persistence of unemployment even when the deficiency of aggregate demand that caused it, no longer exists. [1]

For e.g., a recession causes the long term unemployed to lose their skills and motivation, hence they might not easily obtain work when the economy begins to recover. [1]

[2]

ii) **Structural Deficit**

Structural Deficit is the public sector deficit that would occur if the economy were operating at the potential level of national income or output, i.e., where there is no excess or deficiency of aggregate demand and therefore a zero-output gap. [1]

iii) **Accelerator Coefficient**

The accelerator theory states that investment is determined by the rate of change of national income. If national income rises, then the level of capital [1]

needed to produce more goods and services also rises. This is called induced investment.

The amount of induced investment depends on the change in national income, i.e., $I = \alpha \Delta Y$, where, α , the accelerator coefficient, is the amount by which induced investment depends on national income.

[1]

[2]

iv) J curve effect

The J- curve effect refers to the fact that a devaluation may initially cause a deterioration in the balance of payments before it subsequently improves (once trade has fully adjusted to the new lower exchange rate).

[1]

[6 Marks]

Solution 35: The exchange rate of the Indian Rupees against other currencies is determined by supply and demand.

Demand for Indian Rupees comes from two main sources:

- Overseas firms and consumers buying Indian goods and services. Indian firms will want to be paid in Indian Rupees (or to convert payment into Indian Rupees).
- Overseas investors making investments in India. To buy Indian assets they will need Indian Rupees.

[2]

Similarly supply of Indian Rupees comes from two sources:

- Indian residents purchasing foreign goods and services.
- Indian residents purchasing foreign assets.

[1]

Generally, the Indian Rupees will increase if there is an increase in the demand for Indian Rupees (as US citizens increase their demand for Indian goods, services and assets) and/or a decrease in its supply (as Indian citizens decrease their demands for US goods, services and assets).

[1]

Short run

In the short run speculation is the main determinant of the exchange rate. For example, if RBI increases interest rates, and Indian assets offer higher expected returns than foreign assets, capital (or hot money) will be attracted into India. This is an example of the second source of demand for Indian Rupees given above (foreigners buying Indian Rupees assets). The higher demand will tend to increase the exchange rate. There will also be a lack of supply of Indian Rupees from Indian investors who will decide not to invest abroad when Indian assets offer high returns.

[2]

A speculative increase in the exchange rate makes a subsequent fall more likely since the exchange rate must follow a unique long run path (as described below). Eventually the flow of capital from abroad will be reduced as high returns in Indian Rupees are offset by the expectation of currency losses. Short-run equilibrium is achieved when the exchange rate has risen enough to stop these capital flows.

[2]

Long run

For a given level of output (including the full employment level of output), there is a unique real exchange rate which produces external balance, i.e., a position where the balance on the current account of the balance of payments is just zero.

[1]

In the long run, the nominal exchange rate will move in order to maintain the real exchange rate required to achieve this external balance.

[1]

For example, suppose the India has a high rate of inflation, which has caused a lack of competitiveness and a balance of payments deficit. The lack of demand for Indian Rupees will cause the nominal exchange rate to fall until competitiveness is restored. The nominal exchange rate will have moved onto its purchasing power parity path in order to restore the unique real exchange rate.

Real changes to the economy (e.g. the discovery of oil, or improvements in productivity) might change the real exchange rate needed to produce external balance.

[1]

[Max 8 Marks]

Solution 36: The **bank deposits multiplier** is the number of times greater the expansion of bank deposits is than the additional liquidity in banks that caused it. It is calculated as $b = 1/L$ where L is the liquidity ratio.

[1]

The **money multiplier (m)** is the number of times greater the expansion of money supply is than the expansion of the monetary base that caused it. It is calculated as

[1]

$$m = \Delta M4 / \Delta \text{monetary base}$$

Where

$M4$ is the **Broad money** consisting of cash in circulation plus retail and wholesale bank and building society deposits and

[1]

Monetary base is the narrow money consisting of cash (notes and coins) in circulation

outside the central bank. This is also known as *cash in circulation*.

[1]

The money multiplier typically has a smaller value than the bank deposits multiplier because:

- Some of the extra cash may be withdrawn from the banking system by the nonbank private sector (households and non-bank firms) eg shopkeepers might keep some of the cash received from customers rather than deposit it all in the banks. In this case, the cash isn't redeposited in the banking system and so cannot be used as a basis for further lending. Hence, the money multiplier is less than the bank multiplier.
- Firms and households may wish to borrow less than the banks wish to lend.

[2]

[Max 5 Marks]
